Fairoak Escape Room & Entertainment Venues

Campus LAN Segmentation, Trunking & Spanning Tree - Case Study No. 87

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Project objective: design a segmented three-switch campus LAN for Fairoak using VLANs, 802.1Q trunking, STP, EtherChannel, inter-VLAN routing, ACLs and port security.

1. Business Scenario & Problem

Fairoak Escape Room & Entertainment Venues has three locations: Downtown, Harbor District, and Old Mill. The three sites are interconnected in a triangle so they can share the same puzzle-control and booking platform. The current flat network allows too much traffic to share one broadcast domain and creates a security problem because public guest devices can get close to internal puzzle-control systems.

The switching layer is redesigned using VLAN segmentation, secure 802.1Q trunks, Spanning Tree Protocol and EtherChannel. Inter-VLAN routing is handled centrally so that normal internal VLANs can communicate where required, while the Guest VLAN is blocked from Puzzle-Control and other internal networks.

- Reduce broadcast traffic by separating users into smaller VLANs.
- Use VLAN 999 as the unused native VLAN on trunks.
- Make SW-OldMill the STP root bridge as required by the case study.
- Use an EtherChannel between SW-OldMill and SW-Downtown for redundancy.
- Allow Guest traffic to remain isolated from internal VLANs.
- Use port security on Harbor District guest access ports.

2. Core Concepts Used

VLANs and broadcast domains

A VLAN creates a separate logical broadcast domain. In this project, Staff-Booking, Admin, IT-Mgmt, Puzzle-Control and Guest traffic are separated so that a broadcast from one group does not automatically reach every device in the network.

Why the triangle needs STP

The three switches form a physical triangle. Keeping all three links active at Layer 2 would create a loop. STP keeps the redundancy but blocks one path logically, so frames have a loop-free path while a backup path remains available.

Why EtherChannel is used

The two parallel links between SW-OldMill and SW-Downtown are bundled into one logical Port-channel. This lets both links work together instead of STP treating one of them as a separate redundant path.

Layer 2 and Layer 3 security

Port security limits the number of MAC addresses accepted on a guest access port. The Layer 3 ACL then controls what the Guest VLAN is allowed to reach after traffic reaches the routing switch.

3. VLAN Scheme

The VLAN plan below follows the case study directly. VLAN numbers and IP subnets are separate values, so the Guest VLAN 950 does not require an IP subnet such as 192.168.950.0. The actual screenshots supplied with the project show the 10.10.x.x addressing scheme, including 10.10.28.10, 10.10.18.10 and the 10.10.50.1 gateway.

VLAN ID	Name	Where it lives	Typical devices	Proposed subnet
18	Puzzle-Control	Downtown + Harbor District	Room controllers, sensor panels	10.10.18.0/24
28	Staff-Booking	All locations	Front-desk booking terminals	10.10.28.0/24
38	Admin	Old Mill	HQ finance/admin	10.10.38.0/24
48	IT-Mgmt	All locations	Switch/AP management interfaces	10.10.48.0/24
950	Guest	Harbor District	Public lobby Wi-Fi	10.10.50.0/24
999	Native (unused)	All trunk links	Deliberately unused	No user subnet

4. Proposed Port Assignment

Switch	Access ports	Purpose	Inter-switch links

Switch	Access ports	Purpose	Inter-switch links
SW-OldMill (L3/root)	Fa0/1-3 -> VLAN 28 Fa0/4-6 -> VLAN 38 Fa0/7-8 -> VLAN 48	Staff, Admin, IT management	Gi0/1-2 -> EtherChannel to SW-Downtown Fa0/24 -> SW-Harbor
SW-Downtown	Fa0/1-3 -> VLAN 18 Fa0/4-6 -> VLAN 28 Fa0/7-8 -> VLAN 48	Puzzle-Control, Staff, IT management	Gi0/1-2 -> EtherChannel to SW-OldMill Fa0/24 -> SW-Harbor
SW-Harbor	Fa0/1-3 -> VLAN 18 Fa0/4-5 -> VLAN 28 Fa0/6 -> VLAN 950 Fa0/7-8 -> VLAN 48	Puzzle-Control, Staff, Guest, IT management	Fa0/23 -> SW-Downtown Fa0/24 -> SW-OldMill

5. 802.1Q Trunking & Native VLAN 999

All three inter-switch connections are configured as 802.1Q trunks. VLAN 999 is used as the native VLAN and is intentionally unused. The production VLANs remain tagged across the trunks.

```
! Example trunk configuration - apply on both ends
interface range fa0/23 - 24
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 18,28,38,48,950,999
```

Why VLAN 999 is safer

Native VLAN traffic is sent untagged. Leaving a real production VLAN as the native VLAN can make VLAN hopping attacks easier to attempt. Using an unused VLAN 999 gives the untagged traffic a dead-end VLAN with no normal users or services.

6. Inter-VLAN Routing and Guest Policy

SW-OldMill is proposed as the Layer 3 switch. Each required VLAN receives an SVI that acts as its default gateway. The Guest VLAN is then filtered with an extended ACL.

```
ip routing

interface vlan 18
 ip address 10.10.18.1 255.255.255.0
 no shutdown

interface vlan 28
 ip address 10.10.28.1 255.255.255.0
 no shutdown

interface vlan 38
 ip address 10.10.38.1 255.255.255.0
 no shutdown

interface vlan 48
 ip address 10.10.48.1 255.255.255.0
 no shutdown

interface vlan 950
```

```
 ip address 10.10.50.1 255.255.255.0
 no shutdown

ip access-list extended GUEST-RESTRICT
 deny ip 10.10.50.0 0.0.0.255 10.10.18.0 0.0.0.255
 deny ip 10.10.50.0 0.0.0.255 10.10.28.0 0.0.0.255
 deny ip 10.10.50.0 0.0.0.255 10.10.38.0 0.0.0.255
 deny ip 10.10.50.0 0.0.0.255 10.10.48.0 0.0.0.255
 permit ip any any
```

7. Spanning Tree - Root Bridge and Port Roles

The case study requires SW-OldMill to be the root bridge. A lower STP priority wins the root election, so SW-OldMill is configured with priority 4096. SW-Downtown can be used as the backup root with priority 8192, while SW-Harbor can retain the default priority.

Switch	STP priority	Expected role
SW-OldMill	4096	Root bridge
SW-Downtown	8192	Backup/root candidate
SW-Harbor	32768	Non-root switch

8. EtherChannel - SW-OldMill -> SW-Downtown

The two parallel links between SW-OldMill and SW-Downtown are bundled into one LACP EtherChannel. This provides redundancy and allows both physical links to be used as one logical connection.

```
! On SW-OldMill and SW-Downtown
interface range gig0/1 - 2
 channel-group 1 mode active

interface port-channel 1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 18,28,38,48,950,999
```

After the bundle forms, STP sees Port-channel 1 as one logical link instead of seeing the two physical cables as two separate Layer 2 paths. This prevents STP from unnecessarily blocking one of the parallel links.

```
Switch#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           LACP   Gig0/1(P) Gig0/2(P)
Switch#
```

What should be visible

- Port-channel 1 should exist.
- Both Gigabit links should be members of the channel.
- The EtherChannel should use LACP.
- STP should calculate on Po1 rather than blocking one of the two parallel cables.

9. Port Security - Harbor District Guest VLAN

Harbor District guest access ports are protected with port security. The case requires a maximum of 2 learned MAC addresses per port and violation action set to restrict.

```
interface fa0/6
 switchport mode access
 switchport access vlan 950
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
```

Figure 9.1 - Supplied port-security evidence for FastEthernet0/6.

The supplied output shows Port Security enabled, Port Status secure-up, Violation Mode Restrict, Maximum MAC Addresses 2 and a sticky MAC address. This directly supports the required security configuration.

Why restrict instead of shutdown

Guest areas are public and devices can connect and disconnect frequently. With shutdown, a violation can place the port into an err-disabled state and stop legitimate users until an administrator fixes it. Restrict is more practical here because offending traffic is dropped and the event is counted while the port remains available.

10. VTP Mode Recommendation

Recommended mode: VTP Transparent.

The network contains only three switches and a small number of VLANs. In this situation, manually defining the VLANs on each switch is simple and avoids the risk of an incorrect VTP revision overwriting the VLAN database across the network.

Transparent mode keeps the VLAN database locally controlled on each switch. The trade-off is that VLANs must be created manually on all switches, but the network becomes easier to control in a small lab and reduces the risk of accidental VLAN database changes.

```
Switch#sho vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : FAIROAK
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 00D0.BA55.E800
Configuration last modified by 0.0.0.0 at 3-1-93 00:23:52

Feature VLAN :
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 11
Configuration Revision            : 0
MD5 digest                        : 0x41 0xFE 0x93 0xBA 0x3F 0x13 0xB9 0x7F
                                    0x9A 0xDC 0x09 0x43 0xCA 0xD4 0xCA 0xE3
Switch#
```

11. Link Failure and STP Re-convergence

The required failure test is the link between SW-OldMill and SW-Harbor. Before the failure, the triangle contains one forwarding path and one STP-blocked backup path. When the active link is shut down, STP should move the previously blocked path into forwarding and restore connectivity through the remaining side of the triangle.

```
interface fa0/24
 shutdown
! wait for STP to re-converge
show spanning-tree vlan 18

interface fa0/24
 no shutdown
```

```
Switch#show spanning-tree vlan 18
VLAN0018
  Spanning tree enabled protocol ieee
  Root ID    Priority    4114
             Address     0002.171B.0731
             Cost        22
             Port        24(FastEthernet0/24)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32786  (priority 32768 sys-id-ext 18)
             Address     0040.0B7D.5AA5
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/24           Root FWD 19        128.24   P2p

Switch#
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
```

```
Switch#show spanning-tree vlan 18
VLAN0018
  Spanning tree enabled protocol ieee
  Root ID    Priority    4114
             Address     0002.171B.0731
             Cost        19
             Port        23(FastEthernet0/23)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32786  (priority 32768 sys-id-ext 18)
             Address     0040.0B7D.5AA5
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/23           Root LRN 19        128.23   P2p
Fa0/24           Altn BLK 19        128.24   P2p

Switch#
```

12. Connectivity Verification

12.1 Staff-Booking reachability

```
C:\>ping 10.10.28.10

Pinging 10.10.28.10 with 32 bytes of data:

Reply from 10.10.28.10: bytes=32 time=1ms TTL=127
Reply from 10.10.28.10: bytes=32 time<1ms TTL=127
Reply from 10.10.28.10: bytes=32 time<1ms TTL=127
Reply from 10.10.28.10: bytes=32 time=35ms TTL=127

Ping statistics for 10.10.28.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 35ms, Average = 9ms

C:\>
```

Figure 12.1 - Supplied ping evidence toward 10.10.28.10.

12.2 Puzzle-Control restriction

```
C:\>ping 10.10.18.10

Pinging 10.10.18.10 with 32 bytes of data:

Reply from 10.10.50.1: Destination host unreachable.
Reply from 10.10.50.1: Destination host unreachable.
Reply from 10.10.50.1: Destination host unreachable.
Reply from 10.10.50.1: Destination host unreachable.

Ping statistics for 10.10.18.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

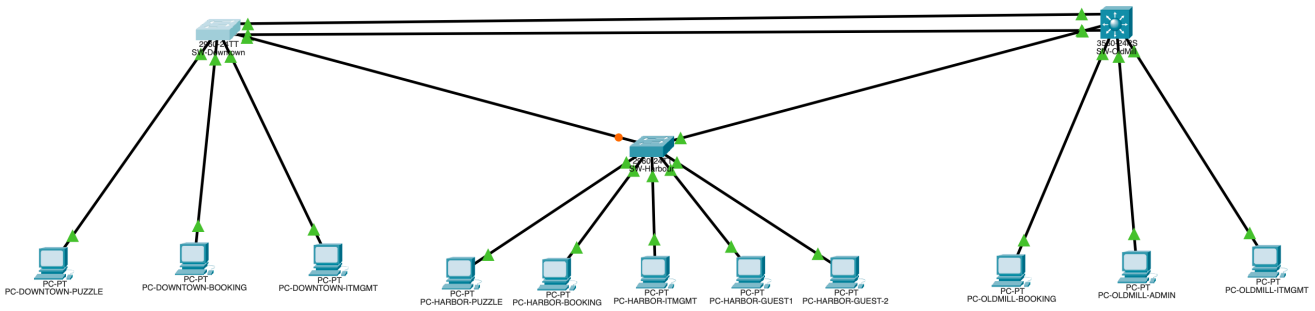
13. Evidence & Screenshots

The screenshots below were supplied with the project. They are placed next to the objective they help prove. Items that were not included in the supplied ZIP are left as clear placeholders rather than inventing evidence.

13.1 Final Packet Tracer topology

13.2 VLAN and port assignments

Final Topology -



Show VLan Brief on all switches

```
Switch#sho vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                 Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                 Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20, Fa0/21, Fa0/22, Fa0/23
18   PUZZLE-CONTROL                    active    Fa0/1
28   STAFF-BOOKING                     active    Fa0/2
38   ADMIN                             active
48   IT-MGMT                           active    Fa0/3
950  GUEST                             active
999  NATIVE-UNUSED                     active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#
```

13.3 Routing table / SVI verification

```
Switch#sho ip int br
Interface              IP-Address      OK? Method Status                Protocol
Port-channel1          unassigned      YES unset  up                    up
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  up                    up
FastEthernet0/5        unassigned      YES unset  up                    up
FastEthernet0/6        unassigned      YES unset  up                    up
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
Vlan1                  unassigned      YES unset  administratively down down
Vlan18                 10.10.18.1      YES manual up                    up
Vlan28                 10.10.28.1      YES manual up                    up
Vlan38                 10.10.38.1      YES manual up                    up
Vlan48                 10.10.48.1      YES manual up                    up
Vlan950                10.10.50.1      YES manual up                    up
Switch#
Switch#
```

```
Switch#sho ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 5 subnets
C       10.10.18.0 is directly connected, Vlan18
C       10.10.28.0 is directly connected, Vlan28
C       10.10.38.0 is directly connected, Vlan38
C       10.10.48.0 is directly connected, Vlan48
C       10.10.50.0 is directly connected, Vlan950

Switch#
```

13.4 MAC address table

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

  18    00d0.9790.bc19    DYNAMIC     Gig0/1
  18    00d0.9790.bc1a    DYNAMIC     Gig0/2
 999    0090.0c5b.5817    DYNAMIC     Fa0/24
 999    0090.0c9e.ed18    DYNAMIC     Po1
Switch#
```

14. ACL, Guest Security and Remaining Evidence

14.1 ACL verification

```
Switch#show access-lists
Extended IP access list BLOCK-GUEST-TO-PUZZLE
    10 deny ip 10.10.50.0 0.0.0.255 10.10.18.0 0.0.0.255 (12 match(es))
    20 permit ip any any (16 match(es))
```

14.3 VTP status

```
Switch#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : FAIROAK
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0090.0C5B.5800
Configuration last modified by 0.0.0.0 at 3-1-93 00:07:17

Feature VLAN :
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 11
Configuration Revision            : 0
MD5 digest                        : 0xD4 0xA6 0x71 0x6D 0xB8 0x29 0x62 0x7E
                                    0x98 0x76 0x88 0xD4 0x0F 0x29 0x02 0xDD
Switch#
```

14.4 EtherChannel summary

```
Switch#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig0/1(P) Gig0/2(P)
Switch#
```

15. Commands Used for Final Verification

```
show vlan brief
show interfaces trunk
show spanning-tree vlan 18
show spanning-tree vlan 28
show etherchannel summary
show ip interface brief
show ip route
```

```
show mac address-table
show ip access-lists GUEST-RESTRICT
show port-security interface fa0/6
show vtp status
```

16. Final Design Summary

The proposed Fairoak network separates the three locations into purpose-based VLANs while keeping the required inter-site triangle. VLAN 18 is reserved for Puzzle-Control at Downtown and Harbor, VLAN 28 is used for Staff-Booking at all locations, VLAN 38 is used for Old Mill Admin, VLAN 48 is used for IT management, and VLAN 950 is used for Harbor guests. VLAN 999 is reserved as the unused native VLAN.

SW-OldMill is selected as the Layer 3 routing switch and STP root. The two Old Mill to Downtown links are bundled into EtherChannel, giving the network redundancy without making STP block one of those parallel links. STP then blocks one side of the remaining triangle until it is needed during a link failure.

The Guest VLAN is protected at two levels. Port security limits the number of learned MAC addresses at the access layer, and the Guest ACL prevents Guest traffic from reaching the internal VLANs, especially Puzzle-Control. The supplied ping and port-security screenshots provide direct evidence for these security and connectivity results.